AQ3.1: Activity Questions 1 - Not Graded

This assignment will not be graded and is only for practice.
Level 1

1 point
Consider the following expression:
$v_1 = (1, 0), v_2 = (0, 0), v_3 = (0, 1), v_5 = 1, v_6 = 5, v_7 = (1, 0, 1), v_8 = (0, 0, 1), v_9 = (0, 0, 0), v_{10} = (0, 1,$
v1 = (1, 0), v2 = (0, 0), v3 = (0, 1), v5 = 1, v6 = 5, v7 = (1, 0, 1), v8 = (0, 0, 1), v9 = (0, 0, 0), v10 =
(0, 1, 0). Choose the set of correct options.

☐
v_1, v_7 represent vectors in R^2 v1, v7 represent vectors in R2.
☐
v_2, v_5 represent vectors in R^1 v2, v5 represent vectors in R1.
☐
v_2, v_5, v_9 represent vectors in R^3 v2, v5, v9 represent vectors in R3.
☐
v_1, v_2, v_3 represent vectors in R^2 v1, v2, v3 represent vectors in R2.
☐
v_5, v_6 represent vectors in R^1 v5, v6 represent vectors in R1.
☐
v_1, v_6 represent vectors in R^1 v1, v6 represent vectors in R1.
☐
v_8, v_9, v_{10} represent vectors in R^3 v8, v9, v10 represent vectors in R3.

No, the answer is incorrect.
Score: 0
Accepted Answers:
v_1, v_2, v_3 represent vectors in R^2 v1, v2, v3 represent vectors in R2.
v_5, v_6 represent vectors in R^1 v5, v6 represent vectors in R1.
v_8, v_9, v_{10} represent vectors in R^3 v8, v9, v10 represent vectors in R3.

1 point
Let v, v_1, v_2 v, v1, v2 represent vectors in a vector space V V over RR. Which of the following expressions make sense and represent vectors in VV ?

☐
$cv \in V$ cv ∈ V , where $c \in R$ c ∈ R.
☐
$\frac{v}{2}$ 2v.
☐
v^2 v2.
☐
$v_1 - v_2$ v1 − v2.
☐
$v_1^2 - v_2^2$ v12 − v22.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$cv \in V$ cv ∈ V , where $c \in R$ c ∈ R.
$\frac{v}{2}$ 2v.

$v_1 - v_2$.

1 point

Consider the set $V = \{(z-x, -y) \mid x, y, z \in R\} \subseteq R^2$ with the usual addition and scalar multiplication as in R^2 and associated statements given below. Choose the correct options.

☐ V is closed under addition i.e $(v_1, v_2 \in V \implies v_1 + v_2 \in V)$.

☐ V is closed under scalar multiplication i.e $(c \in R, v \in V \implies cv \in V)$.

☐ V has a zero element with respect to addition. i.e., there exists some element 0 such that $v + 0 = v$, for all $v \in V$.

☐ V is not a vector space.

☐ $(a+b)v = av + bv$ where $a, b \in R$ and $v \in V$.

No, the answer is incorrect.
Score: 0
Accepted Answers:

V is closed under addition i.e $(v_1, v_2 \in V \implies v_1 + v_2 \in V)$.

V is closed under scalar multiplication i.e $(c \in R, v \in V \implies cv \in V)$.

V has a zero element with respect to addition. i.e., there exists some element 0 such that $v + 0 = v$, for all $v \in V$.

$(a+b)v = av + bv$ where $a, b \in R$ and $v \in V$.

1 point

Consider a set $V = \{(1, x) \mid x \in R\} \subseteq R^2$ with the usual addition and scalar multiplication as in R^2 and associated statements given below. Choose the correct options.

☐ V is closed under addition.

☐ V has no zero element with respect to addition. i.e., there does not exists any element 0 such that $v + 0 = v$, for all $v \in V$.

☐ $(a+b)v = av + bv$ where $a, b \in R$ and $v \in V$.

☐ None of the above.

No, the answer is incorrect.
Score: 0
Accepted Answers:

V has no zero element with respect to addition. i.e., there does not exists any element 0 such that $v + 0 = v$, for all $v \in V$.

$(a+b)v = av + bv$ where $a, b \in R$ and $v \in V$.

1 point

Which of the following sets with the given addition and scalar multiplication (scalars are real numbers in every case) form vector spaces?

$$V_1 = \{(x, y, z) \mid x, y, z \in R\}$$

Addition: $(x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2, y_1 + 1, z_2 + 1)$; $(x_1, y_1, z_1), (x_2, y_2, z_2) \in V_1$

Scalar multiplication: $c(x, y, z) = (x, y, cz)$; $(x, y, z) \in V_1$, $c \in R$

V1 =

{(x, y, z) | x, y, z ∈ R}Addition: (x1, y1, z1) + (x2, y2, z2) = (x1 + x2, y1 + 1, z2 + 1); (x1, y1, z1), (x2, y2, z2) ∈ V1Scalar multiplication: c(x, y, z) = (x, y, cz); (x, y, z) ∈ V1, c ∈ R

$$V_2 = \{(x, y, z) \mid x, y, z \in R\}$$
$$\text{Addition: } (x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1, y_1, z_1); \quad (x_1, y_1, z_1), (x_2, y_2, z_2) \in V_2$$
$$\text{Scalar multiplication: } c(x, y, z) = (x, cy, z); \ (x, y, z) \in V_2, \ c \in R$$
V2 =

{(x, y, z) | x, y, z ∈ R}Addition: (x1, y1, z1) + (x2, y2, z2) = (x1, y1, z1); (x1, y1, z1), (x2, y2, z2) ∈ V2Scalar multiplication: c(x, y, z) = (x, cy, z); (x, y, z) ∈ V2, c ∈ R

$$V_3 = \{(x, y, z) \mid x, y, z \in R\}$$
$$\text{Addition: } (x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2, 0, 0); \quad (x_1, y_1, z_1), (x_2, y_2, z_2) \in V_3$$
$$\text{Scalar multiplication: } c(x, y, z) = (cx, y, z); \ (x, y, z) \in V_3, \ c \in R$$
V3 =

{(x, y, z) | x, y, z ∈ R}Addition: (x1, y1, z1) + (x2, y2, z2) = (x1 + x2, 0, 0); (x1, y1, z1), (x2, y2, z2) ∈ V3Scalar multiplication: c(x, y, z) = (cx, y, z); (x, y, z) ∈ V3, c ∈ R

Choose the correct options.

☐ V_3 V3 is not vector space.

☐ V_2 V2 is vector spaces.

☐ All are vector spaces.

☐ V_1 V1 is not vector space.

No, the answer is incorrect.
Score: 0
Accepted Answers:
V_3 V3 is not vector space.
V_1 V1 is not vector space.

If (a, b, c)(a, b, c) is the inverse element of $(1, 2, 5) \in R^3$ (1, 2, 5) ∈ R3 of the vector space R^3 R3 , then find the value of $a - b + c$a − b + c.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) -4
1 point

Level 2

1 point
Consider the following sets:

- $V_1 = \{A \mid A \in M_{2\times 2}(R)$ and A is a symmetric matrix, i.e., $A = A^T\}$V1 = {A | A ∈ M2×2 (R) and A is a symmetric matrix, i.e., A = AT }

- $V_2 = \{A \mid A \in M_{2\times 2}(R)$ and A is a scalar matrix$\}$V2 = {A ∣ A ∈ M2×2 (R) and A is a scalar matrix}
- $V_3 = \{A \mid A \in M_{2\times 2}(R)$ and A is a diagonal matrix$\}$V3 = {A ∣ A ∈ M2×2 (R) and A is a diagonal matrix}
- $V_4 = \{A \mid A \in M_{2\times 2}(R)$ and A is a upper triangular matrix$\}$V4 = {A ∣ A ∈ M2×2 (R) and A is a upper triangular matrix}
- $V_5 = \{A \mid A \in M_{2\times 2}(R)$ and A is a lower triangular matrix$\}$V5 = {A ∣ A ∈ M2×2 (R) and A is a lower triangular matrix}

Choose the set of correct options.

☐ Only V_1 V1 is a subspace of $M_{2\times 2}(R)$M2×2(R).

☐ Only V_4 V4 is a subspace of $M_{2\times 2}(R)$M2×2(R).

☐ Both V_2 and V_3 V2 and V3 are subspaces of $M_{2\times 2}(R)$M2×2(R)

☐ All are subspaces of $M_{2\times 2}(R)$M2×2(R)

No, the answer is incorrect.
Score: 0
Accepted Answers:
Both V_2 and V_3 V2 and V3 are subspaces of $M_{2\times 2}(R)$M2×2(R)
All are subspaces of $M_{2\times 2}(R)$M2×2(R)

1 point
Choose the set of correct options.

☐ A vector in R^3 R3 can be thought of as matrix of order 1×3 or $3\times 1$1 × 3 or 3 × 1.

☐ Intersection of any two subspaces of a vector space VV is not a subspace.

☐ Intersection of any two subsets of a vector space VV is a subspace.

☐ The empty subset of a vector space VV is a subspace of VV .

☐ $V = \{(x,0,0) \mid x \in R\}$V = {(x, 0, 0) ∣ x ∈ R} is a subspace of R^3 R3.

No, the answer is incorrect.
Score: 0
Accepted Answers:
A vector in R^3 R3 can be thought of as matrix of order 1×3 or $3\times 1$1 × 3 or 3 × 1.
$V = \{(x,0,0) \mid x \in R\}$V = {(x, 0, 0) ∣ x ∈ R} is a subspace of R^3 R3.

1 point
Consider a set $V = \{(x,y) \mid x, y \in R\} \subseteq R^2$ V = {(x, y) ∣ x, y ∈ R} ⊆ R2 with the usual addition as in R^2 R2 and scalar multiplication is defined as

$$c(x,y) = \begin{cases} (0,0) & c = 0 \\ (\frac{cx}{2}, \frac{y}{c}) & c \neq 0 \end{cases} \quad (x,y) \in V,\ c \in R$$c(x, y) = {(0, 0)(2cx, cy)c = 0c = 0

(x, y) ∈ V , c ∈ R

Consider the statements given below.

- P:P: VV is closed under addition.

- Q:Q: VV has zero element with respect to addition. i.e., there exists some element 00 such that $v + 0 = v$v + 0 = v, for all $v \in V$v ∈ V .
- R:R: $1.v = v$1.v = v where $1 \in R$ and $v \in V$1 ∈ R and v ∈ V .
- S:S: $a(v_1 + v_2) = av_1 + av_2$a(v1 + v2) = av1 + av2 where $v_1, v_2 \in V$ and $a \in R$v1, v2 ∈ V and a ∈ R.
- T:T: $(a + b)v = av + bv$(a + b)v = av + bv where $a, b \in R$ and $v \in V$a, b ∈ R and v ∈ V .

Choose the set of correct options.

- [ ] Only P is true.
- [ ] Only Q is true.
- [ ] Both P and Q are true.
- [ ] P,Q and S are true.
- [ ] Both R and T are not true.

No, the answer is incorrect.
Score: 0
Accepted Answers:
Both P and Q are true.
P,Q and S are true.
Both R and T are not true.
Consider the following statements

- Statement P: Statement P: $W = \{A \mid A \in M_{2\times 2}(R)$ and $A^2 = A\}$W = {A | A ∈ M2×2(R) and A2 = A}, a subset of $M_{2\times 2}(R)$M2×2(R) with usual addition and scalar multiplication as in $M_{2\times 2}(R)$M2×2(R), is a subspace of $M_{2\times 2}(R)$M2×2(R).
- Statement Q: Statement Q: $W = \{A \mid A \in M_{2\times 2}(R)$ and $det(A) = 0\}$W = {A | A ∈ M2×2(R) and det(A) = 0}, a subset of $M_{2\times 2}(R)$M2×2(R) with usual addition and scalar multiplication as in $M_{2\times 2}(R)$M2×2(R), is a subspace of $M_{2\times 2}(R)$M2×2(R).
- Statement R: Statement R: Consider a system of linear equations $Ax^T = b$AxT = b, where AA is invertible square matrix of order 2, $x = (x_1, x_2)$x = (x1, x2) and $b \neq 0$b = 0. Let $W = \{x = (x_1, x_2) \mid x \in R^2$ and $Ax^T = b\}$W = {x = (x1, x2) | x ∈ R2 and AxT = b} is a subset of R^2R2 with usual addition and scalar multiplication as in R^2R2, is a subspace of R^2R2.
- Statement S: Statement S: Consider a system of linear equations $Ax^T = 0$AxT = 0, where AA is invertible square matrix of order 2 and $x = (x_1, x_2)$x = (x1, x2). Let $W = \{x = (x_1, x_2) \mid x \in R^2$ and $Ax^T = 0\}$W = {x = (x1, x2) | x ∈ R2 and AxT = 0} is a subset of R^2R2 with usual addition and scalar multiplication as in R^2R2, is a subspace of R^2R2.

Find the number of correct statements

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 1
1 point

Check Answers